

Indus Sahulat

Final Report

By

Team Lead:

Syed Muhammad Irtiza Zaidi - 23973

Members:

Muhammad Hassaan Farooq - 25143

Muhammad Taha Samji - 25186

Shaheer Alam - 25144

Azizullah Khan - 24931

Supervisor:

Dr. Sajjad Haider

Professor

Department of Computer Science

School of Mathematics and Computer Science (SMCS)

Institute of Business Administration (IBA) Karachi

Company Representative:

Syed Ghazanfar Saleem, Zainab Tariq Soomro (POC) - Indus Hospital & Health
Network

Spring Semester 2025

Institute of Business Administration (IBA) Karachi Pakistan

GitHub Links:

<https://github.com/azizullahkhan16/indus-sahulat-backend.git>

<https://github.com/TahaSamji/IndusSahulat-Desktop-Portal.git>

<https://github.com/TahaSamji/Indus-Sahulat-Hospital-portal.git>

<https://github.com/Muhammad-Hassaan-Farooq/Indus-Sahulat-Driver-Portal.git>

https://github.com/Muhammad-Hassaan-Farooq/indus_sahulat.git

Abstract

In underdeveloped countries like Pakistan, the lack of an efficient emergency response system often results in delays in medical assistance, leading to preventable deaths. The "Indus Sahulat" project aims to address this critical issue by developing a mobile application. This application is designed to automate and streamline the emergency ambulance dispatch process for Indus Hospital. The app will enable clients to instantly notify nearby hospitals, prompting the immediate dispatch of ambulances. It will also allow for real-time tracking of the ambulance's location. This system ensures that hospitals are pre-alerted and medical teams are prepared to treat patients upon arrival, thereby significantly reducing wait times and improving emergency care outcomes.

Acknowledgement

We would like to express our profound gratitude to all those who supported and guided us throughout the development of the **Indus Sahulat** project. First and foremost, we extend our heartfelt thanks to **Indus Hospital** for providing us with the opportunity to work on this impactful initiative. Their invaluable input and collaboration played a pivotal role in shaping the app's design and features, while their vision for an efficient and responsive emergency medical system served as our guiding light. We are deeply indebted to our supervisor, **Dr Sajjad Haider**, whose insightful guidance, constructive feedback, and unwavering encouragement were instrumental in helping us navigate challenges and achieve significant milestones. We also acknowledge the support of our batchmates and peers, who shared their knowledge, ideas, and motivation, enhancing the quality of our work through collaboration and camaraderie. Lastly, we are sincerely grateful to our families for their patience, understanding, and encouragement, which enabled us to dedicate ourselves fully to this project. This journey has been an enriching learning experience, and we are immensely thankful to everyone who contributed to the successful completion of this endeavor.

Contributions:

Group Member	Assigned Work
Syed Muhammad Irtiza Zaidi	• Overall App Development Lead• Coordinate and integrate core app features• Oversee Emergency service Feature: One-click emergency activation, response management, hospital capacity monitoring, and ambulance tracking• Integrate an LLM-powered Chatbot for user queries• Implement multi-language support for broader user accessibility
Shaheer Alam	• Develop backend infrastructure and APIs• Implement Ambulance Dispatch feature• Handle Notification System for doctors and Hospital Capacity Monitoring• Ensure robust security integrations and overall app security• Work on Mobile App Frontend
Muhammad Hassaan Farooq	• Develop Patient Interface & Emergency Alert System• Develop the Emergency Alert System for patients• Manage CI/CD Pipelines and deployments• Work on Communication Channels for real-time updates between patients, doctors, and responders
Muhammad Taha Samji	• Medical Records Management• Develop Backend API's• Implement Real-Time Geo Tracking• Implement Medical Records section for accessing history, reports, and prescriptions• Integrate Patient Record Search functionality
Azizullah Khan	• Develop Health Education section with resources• Develop Doctor Profiles feature• Develop Web Server: Real-time crisis management for hospital portals• Database Management• Work on UI/UX design for the application

Table of Contents

Indus Sahulat.....	1
Final Report	1
GitHub Links:	2
Abstract.....	3
Acknowledgement.....	4
Contributions:	5
1. Introduction	8
1.1. Background	8
1.2. Motivation	8
1.3. Problem Statement	9
1.4. Proposed Solution: Indus Sahulat	9
2. Literature Review & Methodology	11
2.1. Literature Review (Market Research & Inspiration)	11
2.2. Methodology	11
3. Software Requirements.....	12
3.1. System Requirements	12
3.1.1. Functional Requirements	12
3.1.2. Non-Functional Requirements.....	13
3.1.3. User Interface Requirements	14
3.2. Functional Requirements Specification	15
3.2.1. Stakeholders	15
3.2.2. Actors and Goals	15
3.2.3. Use Cases	16
3.3. OBE Requirements	18
3.3.1. Ethical Considerations (from Proposal).....	18
3.3.2. Design & Development of Solutions (from Proposal)	18
3.3.3. Detailed OBE Requirements (from SRS)	19
4. Software Design	21
4.1. Interaction Diagrams	21
4.2. Class Diagram and Interface Specification	21
4.3. System Architecture and System Design	23
4.3.1. Architectural Style	23
4.3.2. Identifying Subsystems.....	23
4.3.3. Mapping Subsystems to Hardware.....	24
4.3.4. Persistent Data Storage	24
4.3.5. Network Protocol.....	24
4.3.6. Global Control Flow	25
4.3.7. Hardware/Software Requirements.....	25
4.4. Algorithms & Data Structures.....	26
4.4.1. Emergency Questions Logic.....	26

4.4.2. Google Maps API Integration / OSRM	26
4.4.3. Nearest Ambulance Assignment Algorithm	26
4.4.4. Real-Time Tracking and Location Updates.....	27
4.4.5. Emergency Alert Dispatch System	27
4.4.6. Medical Record Retrieval (Optional Feature)	27
4.4.7. Data Structures Used	28
4.5. User Interface Design & Implementation	28
4.6. Design of Tests	29
5. Conclusion & Future Work.....	31
5.1. Conclusion.....	31
5.2. Progress	31
5.3. Challenges Encountered.....	31
5.4. Future Work/Improvements	32
6. References.....	34

1. Introduction

1.1. Background

The concept of the "Golden Hour" in medicine underscores the critical 60-minute window after a traumatic injury, during which prompt medical treatment significantly improves survival chances and minimizes long-term complications. This principle was popularized by Dr. R. Adams Cowley, who stated, "There is a golden hour between life and death. If you are critically injured, you have less than 60 minutes to survive". In regions like Pakistan, inefficiencies in emergency response systems often lead to delays in medical attention, consequently increasing mortality rates. Existing emergency response systems are often fragmented, leaving hospitals unprepared to manage emergencies effectively in real-time.

1.2. Motivation

The inspiration for the "Indus Sahulat" project stems from research highlighting critical gaps in emergency healthcare, particularly in low-resource settings. A study by Sama Mukhtar et al., "Standing at the edge of mortality; Five-year audit of an emergency department of a tertiary care hospital in a low resource setup," revealed several key insights. The paper reported a mortality incidence of 0.076% (7.6 per 10,000 Emergency Department (ED) visits) over five years. While this figure might seem lower than anticipated, it could be influenced by factors such as the referral of critical patients after initial stabilization due to bed unavailability or lack of specific medical services.

The mean time lapse before presentation was 44 ± 147 hours, and the mean length of stay before death was 3.4 ± 2.8 hours, indicating a need to expedite patient presentation and reduce ED time. Systemic gaps, including limited resources, lack of pre-hospital care, and insufficient physician knowledge at primary/secondary facilities, contribute to delayed presentations and poor outcomes. Furthermore, mortalities were often associated with high triage acuity scores (P1 cases). The research concluded that "Over half the mortalities in low-resource settings may be addressed by improved emergency care systems...encouraging timely presentation to healthcare, availability of pre hospital care and swift transfer to tertiary care may mitigate some deaths". This provided a clear impetus for developing an automated system to streamline emergency services.

1.3. Problem Statement

In Pakistan and similar underdeveloped countries, inefficient emergency response systems are a major concern, often leading to delayed medical attention and, consequently, increased mortality rates. The current process for requesting an ambulance is often inadequate, lacking a streamlined method to alert hospitals and prepare medical teams in advance, which can be fatal in critical emergencies.

The emergency healthcare system grapples with several challenges:

- High mortality rates due to delayed response times.
 - Inefficient resource allocation.
 - Lack of real-time coordinated communication between patients, ambulances, and hospitals.
 - Inadequate preparation of emergency departments for incoming patients.
 - Limited use of modern technology to optimize the process.
 - Fragmented patient information management.
 - Inefficient scheduling of routine services and limited access to health education and preventive care resources further compound the problem.
- These issues collectively contribute to preventable deaths and complications, highlighting an urgent need for an innovative, technology-driven solution.

1.4. Proposed Solution: Indus Sahulat

"Indus Sahulat" is a mobile application designed to address the critical inefficiencies in Pakistan's emergency response systems. The project aims to develop a platform that automates and streamlines the emergency ambulance dispatch process, focusing initially on the Indus Hospital network. The core vision is to transform emergency response and overall healthcare delivery by creating a unified emergency alert system. This system will notify and prepare hospitals and ambulance services in real-time, significantly reducing delays.

Key features of the Indus Sahulat application include:

- **Instant Emergency Alert System:** Allows patients to quickly alert nearby ambulances, automatically sending their pinned GPS location.
- **Real-Time Ambulance Tracking:** Enables patients, hospitals, and drivers to track the ambulance's location live.
- **Pre-Alert System for Hospitals:** Doctors and hospital administrators receive immediate notifications about incoming emergencies, allowing medical teams to be prepared upon the patient's arrival.
- **Role-Specific Interfaces:** The application is designed for three primary roles: patients, doctors, and ambulance drivers, each with tailored functionalities.

- **Integrated Communication:** A messaging system facilitates communication between the patient, ambulance crew, and the hospital.
- **Multi-language Support:** To cater to diverse users in Pakistan.
- **Optional Integrated Health Services:** The platform also aims to address broader healthcare challenges by potentially integrating features for centralized medical records, routine service scheduling, and preventive healthcare education.

By leveraging technology, Indus Sahulat aims to ensure swift coordination, efficient resource allocation during critical emergencies, and a proactive approach to healthcare, ultimately ensuring that individuals receive timely and life-saving care.

2. Literature Review & Methodology

2.1. Literature Review (Market Research & Inspiration)

Several companies and research initiatives are working on automating emergency services to improve response times and efficiency.

- **Thoughtful (thoughtful.ai):** Specializes in automating public safety and emergency response tasks, minimizing manual data entry, reducing errors, tracking calls and incidents, and ensuring compliance with regulations like HIPAA.
- **Unit4 (unit4.com):** Provides software solutions for public sector organizations, offering tools that streamline operations within emergency services and improve data integration and operational efficiency through ERP solutions.
- **RapidSOS (rapidsos.com):** An intelligent safety platform that enhances emergency response by integrating real-time data from over 540 million connected devices to public safety agencies, improving traditional 911 systems. Research has also demonstrated the effectiveness of automated notification systems in public safety. For example, New York City's automated alerting platform, implemented after the 9/11 attacks, now serves over a million subscribers, highlighting the impact of timely automated communications. These innovations align with the goals of Indus Sahulat, which aims to leverage technology for better data management, communication, and response times in emergency medical services. The primary inspiration, as detailed in the Motivation section, comes from the research paper by Sama Mukhtar et al., which identified critical areas for improvement in emergency care systems in low-resource settings.

2.2. Methodology

The Indus Sahulat project follows a structured approach to development, a visual representation of which is outlined in the project's flowchart focusing on user interactions for patients, ambulances, and hospitals. The process involves:

1. **Application Initiation:** Users (Patient, Ambulance, Hospital) open the application leading to a homepage.
2. **User-Specific Paths:**
 - **Patient:** Can trigger an emergency alert, which is then sent to the ambulance system. This involves ambulance dispatch, communication with the driver, ambulance tracking, and confirmation of ambulance arrival.
 - **Ambulance:** Receives dispatch notifications. The driver can accept or decline. If accepted, they track the patient's location, can communicate with the patient, dispatch an alert to the hospital, and confirm arrival.
 - **Hospital:** Receives emergency notifications and patient details. The hospital can accept or decline the patient. If accepted, they can chat with the ambulance crew and prepare for the patient's arrival. This methodology emphasizes real-time notifications, location tracking, and inter-party communication to streamline the emergency response.

3. Software Requirements

3.1. System Requirements

3.1.1. Functional Requirements

The functional requirements for the Indus Sahulat Application are detailed in Table 1.

Table 1: Functional Requirements for Indus Sahulat Application

S.No	Functional Requirement	Description
1	Emergency Alert System	Provides patients with a quick-access button to send emergency alerts. These alerts include the patient's GPS location for precise tracking. Nearby ambulances are automatically notified to enable swift responses.
2	Notification System	Delivers real-time notifications to ambulance drivers for new emergency alerts. Simultaneously, hospital administrators are informed to prepare necessary resources and staff for incoming patients.
3	Ambulance Dispatch	Automates ambulance assignment based on proximity and availability. Enables real-time location tracking of ambulances, visible to patients, hospital administrators, and emergency responders for seamless coordination.
4	Preparation for Arrival	Keeps hospital medical staff updated on the patient's condition, including symptoms and vital information. Estimates ambulance arrival times dynamically, allowing hospitals to be better prepared.
5	Messaging System	Facilitates secure in-app communication between patients and ambulance drivers for efficient coordination. Also supports confidential messaging between ambulance drivers and hospital staff to streamline operations.
6	Multi-language Support	Supports multilingual functionality, initially offering Urdu and English to cater to diverse user groups. Users can select their preferred language during setup or change it later via the settings menu.
7	Health Education	Provides access to a wide range of health and wellness resources, including articles, videos, and instructional materials. Features an intuitive search function for quick access to relevant information.
8	Emergency Condition Assessment	Prompts patients with a series of structured questions when an emergency alert is raised. Collects critical details about the patient's symptoms and medical history. Determines the most suitable ambulance type based on patient inputs to optimize resource allocation.

3.1.2. Non-Functional Requirements

The non-functional requirements for the Indus Sahulat Application are detailed in Table 2.

Table 2: Non-Functional Requirements for Indus Sahulat Application

S.No	Non-Functional Requirement	Description
9	Performance	The system must respond to emergency alerts within 2 seconds to ensure timely action. Live location tracking updates should refresh every 5 seconds to provide real-time information. The application must support at least 5,000 concurrent users without any performance degradation.
10	Scalability	The architecture should support horizontal scaling to accommodate increased load during peak usage. It should maintain optimal performance and response times even as the number of users or emergencies increases. Dynamic load balancing mechanisms should be implemented to ensure scalability.
11	Availability	The system must be operational 24/7 with minimal downtime to support continuous emergency services. Redundant components should be integrated to ensure high availability. A failover mechanism must be in place to switch to backup systems seamlessly in case of a primary system failure.
12	Usability	The application should feature an intuitive and user-friendly interface suitable for users with minimal technical expertise. It must comply with accessibility standards to accommodate users with disabilities. Multi-language support should be implemented seamlessly to cater to diverse user groups.
13	Maintainability	The codebase should be modular and well-documented to facilitate ease of maintenance. The system should support straightforward updates, bug fixes, and feature enhancements. Automated testing and deployment pipelines should be implemented to streamline the release process and ensure high-quality code.
14	Reliability	The system must have a robust backup and recovery plan to handle failures effectively. Redundancy should be implemented at critical points to ensure uninterrupted operations. Emergency alerts must reach at least one responder, even during peak loads or system issues.
15	Security	User data should be encrypted both in transit and at rest to ensure confidentiality. The system must comply with applicable data protection regulations such as GDPR to safeguard user privacy. Strong authentication mechanisms

		and regular security audits should be employed to protect the system from vulnerabilities.
16	Compatibility	The application should be fully compatible with both iOS and Android platforms, supporting a wide range of devices. It must integrate seamlessly with existing hospital systems to enhance interoperability. The app should also adapt to various screen sizes and resolutions to provide a consistent user experience across devices.

3.1.3. User Interface Requirements

The user interface (UI) for Indus Sahulat is designed for simplicity and accessibility, considering diverse users in Pakistan, including those with minimal technical expertise.

- **General Design Principles:**
 - A clean and intuitive interface with a minimal learning curve.
 - Familiar and culturally relevant icons, buttons, and layouts for easy navigation.
 - Consistent use of color schemes and fonts to enhance readability and usability.
- **Accessibility:**
 - Multi-language support for Urdu and English, with clear language selection options.
 - Readable fonts, high-contrast themes, and voice-over support for users with visual impairments.
 - Easy-to-use interface for individuals with limited exposure to technology.
- **Responsiveness:**
 - Compatibility across a wide range of devices, including low-cost smartphones.
 - Adaptability to various screen sizes and resolutions.
- **Core Features by Role:**
 - **Patients:**
 - Prominently placed emergency alert button for quick access.
 - Real-time ambulance tracking with simple visual representation and ETA.
 - **Ambulance Drivers:**
 - Clear and straightforward navigation tools with step-by-step directions.
 - Notifications with audible alerts for immediate response.
 - **Hospital Administrators:**
 - Concise and organized dashboard to monitor emergencies, track ambulances, and manage resources.
- **Visual Feedback:**
 - Real-time updates on actions like successful alert submissions or arrival estimates.
 - Informative prompts and confirmation messages to guide users.
- **Security and Privacy:**
 - Secure login options, potentially including biometric authentication.

- Privacy measures to safeguard sensitive information, aligning with local and international standards.
- **Error Handling and Recovery:**
 - User-friendly error messages with actionable suggestions.
 - Backup options to retry failed operations seamlessly. The interface aims for inclusivity across urban/rural populations, age groups, and literacy levels. Examples of screens include Login/SignUp, Searching Ambulance screens, and Searching Hospital screens.

3.2. Functional Requirements Specification

3.2.1. Stakeholders

The key stakeholders for the Indus Sahulat system are:

1. **Patients/General Public:** Individuals requiring urgent medical attention (primary focus) or routine healthcare services. They use the system to send emergency alerts, share location, and potentially access health records and wellness resources.
2. **Emergency Responders:** Including ambulance drivers and paramedics. They receive real-time alerts, navigation, and patient information to provide timely medical care and transport.
3. **Hospital Administrators:** Responsible for coordinating resources and ensuring hospital readiness for incoming emergencies. They receive real-time updates to manage patient care and hospital resources efficiently.
4. **Emergency Healthcare Professionals:** ER doctors, nurses, and specialists dedicated to critical care.
5. **Healthcare Policymakers:** Officials involved in improving emergency healthcare systems.
6. **System Administrators:** Responsible for maintaining system functionality, security, monitoring, and managing user access.
7. **Super Admin / Ambulance Provider (e.g., Edhi, Chippa):** Oversee ambulance services and different types of ambulances (emergency responsive, medical assisted, patient carrying).
8. **Hospitals (e.g., Indus Hospital):** Entities that receive patients and manage their care.

3.2.2. Actors and Goals

1. **Patients:**
 - **Goal:** Receive timely emergency responses and access health resources.
 - **Responsibilities:** Initiate emergency alerts, provide medical information (if possible), and share location.
2. **Ambulance Drivers:**
 - **Goal:** Respond promptly to emergencies and transport patients safely.
 - **Responsibilities:** Follow dispatch instructions, navigate to patient locations, and communicate updates.
3. **Hospital Administrators/Staff:**
 - **Goal:** Ensure hospital readiness, efficient resource management, and prepare for patient arrival.

- **Responsibilities:** Coordinate hospital resources, manage staff, monitor system updates, and view patient/incident information.
- 4. **System Administrators:**
 - **Goal:** Maintain system functionality and security.
 - **Responsibilities:** Monitor and maintain the system, ensure security, and manage user access.

3.2.3. Use Cases

The interactions of different actors with the Indus Sahulat app are defined by several use cases. Detailed descriptions are provided in subsequent tables.

Table 3: Initiate Emergency Alert Use Case

Field	Description
Use Case	Initiate Emergency Alert
Description	A patient initiates an emergency alert through the app.
Actors	Patient, System
Preconditions	The patient is logged into the app.
Postconditions	An ambulance is dispatched, and notifications are sent to the hospital and ambulance driver.
Main Flow	1. Patient opens the app and selects the emergency alert button. 2. The system asks for emergency details. 3. Patient provides answers, and the system determines the ambulance category. 4. The system dispatches the ambulance with location and details, notifying the hospital and driver.

Table 4: Dispatch Ambulance Use Case

Field	Description
Use Case	Dispatch Ambulance
Description	The system automatically dispatches the appropriate ambulance.
Actors	System, Ambulance Driver
Preconditions	An emergency alert has been initiated.
Postconditions	The ambulance is in route.
Main Flow	1. The system receives the emergency alert. 2. It selects the nearest appropriate ambulance. 3. Location and details are sent to the driver, who acknowledges and starts the journey. 4. The system updates the ambulance status.

Table 5: Track Ambulance Progress Use Case

Field	Description
Use Case	Track Ambulance Progress
Description	The patient and hospital can track the real-time progress of the ambulance.

Actors	Patient, Hospital, System
Preconditions	An ambulance has been dispatched.
Postconditions	The patient and hospital know the ambulance's progress.
Main Flow	1. The patient and hospital open the tracking feature. 2. The system shows real-time location and updates (e.g., every 10 seconds). 3. The system shows the ETA.

Table 6: Communicate via Messaging System Use Case

Field	Description
Use Case	Communicate via Messaging System
Description	Patients, ambulance drivers, and hospitals can communicate via messaging.
Actors	Patient, Ambulance Driver, Hospital, System
Preconditions	The actors are logged in.
Postconditions	Messages are exchanged successfully.
Main Flow	1. The actor opens the messaging feature. 2. They compose a message and select the recipient. 3. The system sends the message, and the recipient can reply. 4. The system supports group messaging for coordination.

Table 7: Access Health Education Resources Use Case

Field	Description
Use Case	Access Health Education Resources
Description	Users can access health resources through the app.
Actors	User, System
Preconditions	The user is logged into the app.
Postconditions	The user has accessed health content.
Main Flow	1. The user opens the app and navigates to health education. 2. The user searches or browses available resources. 3. The user selects a resource to view. 4. The system displays the resource (article, video, etc.).

Other implicit use cases include:

- **Monitor Hospital Capacity:** Hospitals (and the system) can monitor capacity to ensure patients are sent to appropriate facilities.
- **Access Medical Records:** Patients can access their medical history.

3.3. OBE Requirements

The development and deployment of the Indus Sahulat app are guided by Outcome-Based Education (OBE) principles, emphasizing ethical considerations, design and development solutions, and overall societal impact.

3.3.1. Ethical Considerations (from Proposal)

- **Patient Privacy and Data Security:** All patient data must be encrypted and stored securely, complying with regulations like HIPAA and GDPR. Anonymizing data where possible is also a safeguard.
- **Informed Consent:** Users must be clearly informed about data collection, use, and sharing, with explicit consent obtained. Users should have opt-out options.
- **Accuracy and Reliability:** Algorithms require regular audits for bias, and app recommendations need continuous validation through testing and peer review. Transparency about capabilities and limitations is crucial.
- **Emergency Response Protocols:** Designed for timely and accurate assistance, with backup systems and human oversight for critical situations.
- **Equity and Accessibility:** Support for multiple languages is essential for diverse populations.
- **User Education and Support:** Comprehensive training, support, and a feedback mechanism are necessary. Regular updates based on feedback will maintain relevance.
- **Ethical Decision-Making:** An ethical review board should oversee development, with transparency in processes and stakeholder involvement.
- **Responsibility and Accountability:** Clearly defined roles for development, maintenance, and oversight, with protocols for liability. Continuous monitoring and addressing of ethical issues are required.

3.3.2. Design & Development of Solutions (from Proposal)

The development will analyze existing automated emergency service products like thoughtful.ai, unit4.com, and rapidsos.com to guide the app's design.

- **Legal Considerations:** Compliance with data protection laws (HIPAA, GDPR) and medical regulations is paramount. Legal experts should be consulted.
- **Environmental Considerations:** Design for compatibility with a wide range of devices to reduce hardware upgrades and promote sustainability.
- **Public Health and Safety Considerations:** The app must provide timely and accurate assistance, with rigorous testing and validation of algorithms, and maintained human oversight.

- **Cultural Considerations:** Cultural sensitivity is vital, ensuring the app is inclusive of diverse healthcare practices and preferences, supporting multiple languages and cultural adaptations.
- **Societal Issues:** Address the digital divide and accessibility for users, including those with minor disabilities, and offer alternative access means to promote equity.

3.3.3. Detailed OBE Requirements (from SRS)

The Indus Sahulat app is designed to connect patients, doctors, and ambulance drivers in real-time, aligning with ethical principles, societal values, and legal standards.

1. **Inclusivity and Accessibility:**
 - **Ethical Actions:** Incorporate features like screen readers, voice commands, customizable fonts; design for low-connectivity environments; ensure compatibility across devices.
 - **Potential Risks:** Exclusion of users in underserved regions or those with disabilities; untimely assistance due to lack of offline capabilities.
2. **Privacy and Security of Health Data:**
 - **Ethical Actions:** Encrypt all health data; obtain clear informed consent; implement strict access controls and authentication; comply with HIPAA and GDPR.
 - **Potential Risks:** Data breaches compromising sensitive information; legal consequences from non-compliance.
3. **Algorithmic Bias and Fairness in Emergency Response:**
 - **Ethical Actions:** Conduct regular algorithm audits for medical urgency-based prioritization; test across diverse socio-economic/geographical groups.
 - **Potential Risks:** Favoring wealthier/urban users, leading to unequal access; exacerbating existing health disparities.
4. **Informed Consent and User Autonomy:**
 - **Ethical Actions:** Provide clear consent forms; allow users to manage/delete data and withdraw consent; implement mechanisms for consent if user is incapacitated.
 - **Potential Risks:** Users not fully understanding consent implications; difficulty obtaining consent in emergencies for incapacitated users.
5. **Dual-Use Risks and Misuse of Technology:**
 - **Ethical Actions:** Implement robust security (e.g., multi-factor authentication); monitor for suspicious activity and have reporting processes; consider and prevent dual-use concerns.
 - **Potential Risks:** Malicious use of tracking/communication features (stalking, fraud); misleading responders or creating false emergencies.
6. **Transparency and Accountability:**
 - **Ethical Actions:** Provide clear information on functionality, data usage, and risks; establish mechanisms for reporting issues and ensure prompt addressing; conduct regular performance audits.
 - **Potential Risks:** User distrust due to lack of transparency; unresolved functionality issues affecting service quality.
7. **Cultural Sensitivity and Global Applicability:**

- **Ethical Actions:** Include multilingual support and culturally relevant health information; conduct user testing with diverse groups.
- **Potential Risks:** Miscommunication or confusion in emergencies due to cultural insensitivity; inadequate support for non-native speakers hindering understanding of critical instructions. Addressing these OBE requirements proactively is essential for the app's success and trustworthiness.

4. Software Design

4.1. Interaction Diagrams

The interaction diagrams illustrate the sequence of operations for key functionalities within the Indus Sahulat system.

- **Initiate Emergency Alert:**
 1. Patient opens the app and selects the emergency alert button.
 2. The system asks for emergency details.
 3. Patient provides answers.
 4. The system determines the appropriate ambulance and makes a request to the AmbulanceProvider.
 5. Ambulance is dispatched.
- **Communicate via Messaging System:**
 1. Patient composes a message and selects the recipient (Ambulance Driver).
 2. The system sends the message to the selected recipient.
 3. Ambulance Driver writes a reply to the patient.
 4. The system sends the reply to the patient.
- **Track Ambulance Progress:**
 1. Patient opens the tracking feature.
 2. The system shows the real-time location of the ambulance.
 3. The system shows the ETA (Estimated Time of Arrival).
- **Access Health Education Resources:**
 1. Patient navigates to health education within the app.
 2. Patient selects a resource (e.g., article, video).
 3. The system shows the selected resource.
 4. The user is navigated to the particular resource content.

4.2. Class Diagram and Interface Specification

The UML Class Diagram depicts the main entities in the Indus Sahulat system and their relationships. Key classes include:

- **User:** Represents individuals interacting with the system (Patient, Doctor, Driver, Admin) with attributes like `userId`, `name`, `email`, `phoneNumber`, `role`, `CNIC` and methods like `login()`, `logout()`.
- **Patient:** A specialization of `User`, with `medicalHistory` and methods like `sendEmergency()`, `trackAmbulance()`.
- **EmergencyAlert:** Contains details of an emergency, such as `alertId`, `patientId`, `timestamp`, `severityLevel`, `status`, `location`, and methods `createAlert()`, `updateStatus()`.
- **Ambulance:** Represents an ambulance vehicle with `currentLocation`, `status`, and method `updateLocation()`. It is associated with an `AmbulanceDepartment`.
- **AmbulanceDepartment:** Manages ambulances with `departmentId`, `location`, and methods `dispatchAmbulance()`, `monitorAmbulance()`, `assignAmbulance()`.
- **Hospital:** Represents a healthcare facility with `hospitalId`, `name`, `location`, `phoneNumber`, `role` (likely admin role within hospital context), and methods like `receivePatient()`.
- **Location:** Stores geographical coordinates (`latitude`, `longitude`).

- **Message:** Facilitates communication with `messageId`, `senderId`, `receiverId`, `content`, `timestamp`, and methods `sendMessage()`, `receiveMessage()`.

Relationships:

- A Patient (User) can have zero or one EmergencyAlert.
- An EmergencyAlert is associated with one Location.
- An AmbulanceDepartment can have one or more Ambulances and is associated with one Location.
- A Hospital is associated with one Location.
- Users can send/receive Messages.

4.3. System Architecture and System Design

4.3.1. Architectural Style

The Indus Sahulat system employs a Client-Server Architecture with a Mobile-Backend Framework.

- **Frontend:** A Flutter mobile application serves as the client for patients, doctors, and ambulance drivers. A Web Portal also exists, likely for hospital admin staff and super admin roles.
- **Backend:** Developed using Spring Boot, it acts as the core system, handling business logic, data storage, and communication. The presentation also mentions Supabase with SQL, possibly for database services, and a load balancer.
- **Database:** A relational database system (PostgreSQL) is used for data management. The presentation also mentions Supabase (SQL) which could be PostgreSQL, with Master/Slave DB replication for read/write operations and caching (Redis).
- **Real-time Communication:** The system integrates real-time communication using WebSockets or Firebase for instant notifications and tracking.

4.3.2. Identifying Subsystems

The system is composed of several interconnected subsystems:

- **User Management Subsystem:** Handles user authentication (JWT-based) and roles for patients, doctors, and ambulance drivers.
- **Emergency Alert and Dispatch Subsystem:** Manages emergency alerts initiated by patients. It finds the nearest suitable ambulance and assigns it to the emergency. It also notifies hospitals and doctors about incoming patients.
- **Real-Time Tracking Subsystem (Geo-Tracking Service):** Uses GPS to track ambulances in real time. It sends live location updates to patients and hospitals and can calculate the shortest path, integrating with services like Google Maps API or OSRM.
- **Communication Subsystem (Notification System):** Facilitates instant messaging between ambulance drivers, patients, and doctors/hospitals. It supports push notifications for real-time updates to all roles.
- **Hospital Admin Panel Subsystem (Hospital Dashboard):** Allows hospital administrators to monitor ambulance dispatch, view incoming patient details, update resource availability, and manage overall emergency response from the hospital's end.
- **Data Management Subsystem:** Responsible for storing medical records, ensuring data security, and managing user profiles and other critical data like ambulance and hospital information.
- **Admin Dashboard (Super Admin):** Provides system analytics and user access management capabilities.

4.3.3. Mapping Subsystems to Hardware

The Indus Sahulat system, being client-server based, distributes its subsystems across various hardware components as shown in Table 8. The presentation system architecture also illustrates a cloud-hosted environment.

Table 8: Mapping of Subsystems to Hardware

Subsystem	Runs On
Flutter Mobile App	Patient, Doctor, Ambulance Mobile Devices
Spring Boot Backend	Cloud Server (e.g., AWS, GCP, Azure)
Database (PostgreSQL/SQL)	Cloud Database Server (e.g., Supabase, Azure SQL)
Real-Time Tracking	Firebase / WebSocket Server / OSRM
Hospital Admin Portal	Hospital Computers (Browser-based)
Web Portal (Super Admin)	Admin Computers (Browser-based)

4.3.4. Persistent Data Storage

The system utilizes a relational database, likely PostgreSQL, for persistent data storage. The conceptual database schema and the architecture diagram illustrate key entities:

- **Patients:** Stores patient information and medical history.
- **Ambulance Driver / Ambulance:** Information about drivers and ambulance vehicles, including their status and location.
- **Hospital / Doctors:** Details about hospitals, their capacity, and associated medical staff.
- **Emergency Request / Incident Events:** Data related to emergency alerts, their status, associated patient, and assigned ambulance.
- **Location:** Stores geo-coordinates for patients, ambulances, and hospitals.
- **Medical History:** Linked to patients. The database design supports replication with a Master DB for write operations and Slave DBs for read operations to enhance performance and availability, along with caching mechanisms like Redis.

4.3.5. Network Protocol

Communication within the Indus Sahulat system relies on standard network protocols as outlined in Table 9.

Table 9: Network Protocols Used in the System

Feature	Protocol
API Communication	REST over HTTPS
Real-Time Tracking	WebSockets / Firebase
Push Notifications	Firebase Cloud Messaging (FCM)
Database Communication	JDBC over TLS

Authentication	OAuth2/JWT
Routing (OSRM)	HTTP API

4.3.6. Global Control Flow

The system operates on a hybrid execution model:

- **Linear Execution:** Standard REST API calls for user authentication, fetching data, and performing actions like updating profiles follow a linear request-response pattern.
- **Event-Driven Execution:** Core functionalities such as emergency alerts, real-time ambulance tracking, and push notifications are event-driven. An alert from a patient triggers a cascade of events including ambulance dispatch and hospital notification. Location updates from ambulances are events that update the map views for patients and hospitals.
- **Time Dependency:**
 - **Timers/Schedulers:** Periodic location updates are sent from ambulances at set intervals. ETAs are dynamically recalculated based on real-time data.
- **Concurrency Management:**
 - The Spring Boot backend is designed to handle multiple concurrent API requests using multi-threading.
 - Database transactions are employed to ensure atomicity and data consistency, for example, in preventing double booking of ambulances or managing patient status updates.
 - WebSocket connections manage numerous simultaneous real-time tracking events and messaging sessions. The state diagram (Figure 19) further illustrates the flow of control based on user actions and system events, covering patient requests, provider assignments, driver actions, and hospital responses.

4.3.7. Hardware/Software Requirements

The minimum and recommended hardware and software requirements for various components of the Indus Sahulat system are specified in Table 10.

Table 10: Hardware and Software Requirements

Component	Minimum Requirement
Client-Side (Mobile App)	
Hardware	Android 8.0+ / iOS 12+, 2GB RAM, GPS-enabled device
Software	Flutter 3.0+, Firebase SDK
Server-Side (Backend and Database)	
Hardware	4 vCPUs, 8GB RAM, 50GB SSD Storage (Cloud-based)
Software	Ubuntu 20.04 LTS, Java 17, Spring Boot, PostgreSQL, Nginx/Apache, Azure services, OSRM

Hospital Admin Portal (Client)	
Hardware	Standard desktop/laptop computer
Software	Modern Web Browser (Chrome, Firefox, Edge)

4.4. Algorithms & Data Structures

4.4.1. Emergency Questions Logic

When an emergency alert is triggered, a set of predefined, structured questions is presented to the patient or a bystander to gather crucial information about the situation. This information helps in assessing the patient's condition and preparing the responding medical team.

The questions aim to identify:

- Known medical conditions (e.g., Diabetes, heart disease, allergies).
- Patient's consciousness and breathing status (Yes/No).
- Current symptoms or condition (e.g., Unresponsive, severe bleeding, chest pain, difficulty breathing).
- Nature of the emergency (e.g., Accident, heart attack, stroke, injury).
- Need for additional services (e.g., Neonatal, trauma care). The collected information is immediately relayed to the ambulance crew and hospital staff to facilitate rapid, informed decision-making and determine the most suitable ambulance type. A priority algorithm uses this information for incident importance and ambulance classification.

4.4.2. Google Maps API Integration / OSRM

Google Maps API (or similar services like OSRM - Open Source Routing Machine) is integral for geospatial capabilities.

- **Real-Time Tracking:** Enables real-time tracking of ambulances, allowing patients and hospital administrators to monitor ambulance locations on an interactive map.
- **Route Optimization:** Calculates the most efficient paths for ambulances to the patient and then to the hospital, considering live traffic conditions, road closures, and other dynamic factors. Ambulance drivers receive turn-by-turn navigation to minimize travel time. Patients can visualize the ambulance's live movement. Figure 7 shows an example of shortest path calculation.

4.4.3. Nearest Ambulance Assignment Algorithm

- **Purpose:** To assign the closest available and appropriate ambulance to an emergency request.
- **Algorithm Type:** Utilizes Dijkstra's Algorithm principles or the Haversine Formula for distance calculation.
- **Steps:**
 1. Receive Emergency Alert with the patient's GPS location.
 2. Fetch GPS coordinates of available ambulances from the database.

3. Calculate the distance between the patient and each available ambulance using the Haversine formula.
4. Consider ambulance type/capabilities based on emergency condition assessment.
5. Select the ambulance with the shortest distance and appropriate capabilities, factoring in estimated response time.
6. Notify the selected ambulance and the nearest (or selected appropriate) hospital.

4.4.4. Real-Time Tracking and Location Updates

- **Purpose:** To continuously update and broadcast the ambulance's location for tracking by patients and hospitals.
- **Algorithm Type:** WebSocket-based Real-Time Communication or Firebase Realtime Database.
- **Steps:**
 1. Ambulance's GPS-enabled device sends location updates periodically (e.g., every 5-10 seconds) to the backend server.
 2. Backend stores these updates, potentially in a temporary fast-access store or directly updates the relevant incident record.
 3. The WebSocket server or Firebase broadcasts the updated location data to the connected clients (patient's app, hospital dashboard) in real time.
 4. A fallback mechanism (e.g., polling) might be used if WebSocket connection is lost, though this is less efficient.

4.4.5. Emergency Alert Dispatch System

- **Purpose:** To automatically process and route emergency requests to the nearest appropriate hospital and assign an ambulance.
- **Algorithm Type:** Priority Queue / Event-Driven Messaging.
- **Steps:**
 1. When a patient sends an alert, the system creates an incident/event and may add it to a priority queue based on initial assessment or severity.
 2. Alerts with higher severity (e.g., cardiac arrest, severe trauma as determined by emergency questions) are prioritized.
 3. The system triggers the Nearest Ambulance Assignment Algorithm.
 4. An event-driven message queue (e.g., Kafka/RabbitMQ, or simpler notification service for this scale) ensures asynchronous communication and fast response to notify all relevant parties (ambulance, hospital).

4.4.6. Medical Record Retrieval (Optional Feature)

- **Purpose:** To allow authorized users (e.g., patients, and with consent, doctors) to retrieve patient medical history securely and efficiently.
- **Algorithm Type:** Indexed Hashing or standard database indexing for fast lookup.
- **Steps:**

1. Patient's medical record ID (or other unique identifiers) is used for lookup.
2. The system queries the indexed database (e.g., PostgreSQL B-Tree Index on patient ID or medical record number).
3. Relevant medical records are retrieved efficiently without requiring full-table scans, ensuring quick access during emergencies or consultations. Data access is governed by strict permissions.

4.4.7. Data Structures Used

- **Hash Tables (or equivalent in database indexing):** For fast lookup of patient records, ambulance status, hospital details. Allows $O(1)$ average time complexity for fetching data by ID.
- **Priority Queue:** Potentially used for managing incoming emergency alerts, prioritizing them based on severity level derived from the emergency condition assessment. Critical emergencies receive higher priority for ambulance dispatch.
- **Graph (conceptual for route optimization):** Road networks can be represented as graphs where intersections are nodes and roads are edges. Dijkstra's Algorithm or A* search algorithm (often internal to mapping services like Google Maps/OSRM) is used to find the shortest/fastest route.
- **Linked Lists (or dynamic arrays/lists for message queues):** Can be used for managing messages in the real-time chat system between patients, ambulances, and hospitals, ensuring messages are processed in order.
- **B-Trees (standard in relational databases like PostgreSQL):** Medical records, user data, and other critical information are stored with B-Tree indexes on key columns, allowing for fast searching, sorting, and retrieval operations.

4.5. User Interface Design & Implementation

The user interface of Indus Sahulat is designed to be intuitive and accessible across various roles and devices.

- **Login/Signup Screens:** Standard screens for user authentication, allowing users to log in with credentials or sign up for a new account.
- **Emergency Screens / Searching Ambulance Screen:**
 - A prominent "CALL AMBULANCE" button is available.
 - Screens show the status, such as "Finding you an ambulance" with a progress indicator, and "Ambulance Arrived".
 - Post-ambulance arrival, a "FINDING NEARBY HOSPITAL" status is shown.
- **Searching Hospital Screens:**
 - Displays a map with hospital locations pinned.
 - A search bar allows users to find hospitals.
 - A list view shows nearby hospitals with details and an option to "SELECT HOSPITAL".
 - A map view shows the route to the selected hospital.
- **Questionnaire Screens:** Screens to collect crucial information during an emergency alert, as described in the Emergency Questions Logic section.
- **General App Functionalities (Optional) Screens:**

- **Doctor Profiles:** May include views for doctor specializations, ratings, and availability.
 - **Medical Records:** Interfaces to access and manage personal medical history, upload/download reports.
 - **Health Education:** Sections for articles, videos, and health forums.
- The overall UI aims for clarity, ease of navigation, and responsiveness, particularly for users with minimal technical expertise and on low-cost smartphones.

4.6. Design of Tests

A comprehensive set of test cases has been designed to ensure the functionality, reliability, and performance of the Indus Sahulat system. Key test areas are summarized in Table 11. The FYP presentation also outlines test cases and API collections used for testing.

Table 11: Design of Tests for Indus Sahulat (Selected Examples)

S.No	Test Case Description	Expected Output
1	Trigger Emergency Alert	Alert is created, patient location recorded, notifications sent.
2	Validate Emergency Alert Data (Missing Data)	System rejects request with an error message.
3	Emergency Alert Status Update (Pending, Accepted etc.)	Database reflects the updated status.
4	Assign Nearest Ambulance	System assigns the closest, appropriate, available ambulance.
5	Track Ambulance Location (Real-time)	Live location updates correctly on patient's/hospital's UI.
6	Update Ambulance Status (Available, In Transit etc.)	Status updates correctly in the system.
7	Notify Hospital on Emergency	Hospital administrators receive an emergency notification.
8	Verify Hospital Capacity Handling (If full)	System redirects to the next nearest available hospital or indicates no capacity.
9	User Login Authentication (Valid/Invalid)	Successful login with valid credentials; error message otherwise.
10	Role-Based Access Control	Access denied if an unauthorized user attempts restricted features.
11	Send and Receive Messages (Real-time chat)	Message is sent and received instantly by intended recipients.
12	Find Nearest Hospital (with capacity)	System returns the closest hospital with available capacity for the emergency.
13	Calculate Shortest Route (via Google Maps/OSRM)	The shortest and fastest route is returned and displayed.

Additional test categories would include performance testing (concurrent users, response times), security testing (data protection, vulnerability checks), usability testing across different user roles, and compatibility testing on various devices and platforms.

5. Conclusion & Future Work

5.1. Conclusion

The "Indus Sahulat" project endeavors to significantly enhance the emergency medical response system in Pakistan, a region where timely medical intervention is often hampered by systemic inefficiencies. By developing a comprehensive mobile application, the project aims to streamline the process from emergency alert generation to patient arrival at a prepared medical facility. The system focuses on leveraging technology to facilitate instant communication, real-time tracking, and efficient coordination among patients, ambulance services, and hospitals. The ultimate goal is to reduce delays, improve resource allocation, and thereby save lives by making the "Golden Hour" more achievable for critically injured or ill individuals. The platform's design, inspired by identified needs and supported by market research, intends to create a more responsive, efficient, and proactive emergency care ecosystem, particularly within the Indus Hospital network initially.

5.2. Progress

The project has made significant progress across its core and optional functionalities as of the last update.

Core Features:

- Emergency Alert System: Completed.
- Notification System: Completed.
- Ambulance Dispatch: Completed.
- Real-Time Tracking: Completed.
- Preparation for Arrival: Completed.
- Messaging System: Sockets applied, indicating completion or near completion.
- Multi-language Support: Completed.

Optional Functionalities:

- Medical Records: Awaiting API from Indus Hospital for integration.
- Health Education: Resources to be provided for integration.

5.3. Challenges Encountered

The development and testing phases of the Indus Sahulat project faced several challenges:

- **Testing Challenges:**
 - **Complex Testing Process:** Required using multiple working portals simultaneously, making testing time-consuming and prone to errors, including concurrency issues.
 - **Consistency Across Portals:** Local changes frequently impacted other portals, complicating efforts to maintain system-wide consistency.

- **Limitations of Free Services:** Use of free services for components like mapping or notifications introduced errors, slower response times, and limited reliability, hindering progress.
- **Time-Consuming Setup:** Establishing a fully integrated and functional project environment took significant time, delaying the development process.
- **Resource Challenges:**
 - **Limited VM Resources:** The deployed virtual machine had low RAM, and usage incurred additional costs, limiting scalability and performance during development and testing.
 - **Map Service Limitations:** The open-source map solution (OSRM), while free, was not fully optimized, leading to lower quality in routing accuracy and map visuals compared to premium services.
 - **Message Broker Overhead:** Implementing a message broker for real-time notifications added complexity.
 - **Verification & SMS Services:** Reliance on third-party services for OTP and authentication (e.g., SMS) could incur costs and dependencies.
- **Domain Challenges:**
 - **Medical Questionnaires:** The medical screening questions required further refinement for accurate emergency assessment and response prioritization.
 - **Ambulance Categorization:** Lack of real-world data made it difficult to effectively categorize ambulances based on capabilities and availability for optimal dispatch.
 - **Hospital Categorization:** Integration of real-world data was needed to simulate accurate hospital filtering based on specialties and capacity.
 - **Classification Logic:** The logic for incident classification based on user questionnaire responses needed improvement for better automated decision-making.

5.4. Future Work/Improvements

Based on the progress and challenges, the following improvements and future enhancements are planned for Indus Sahulat:

- **UI/UX Enhancements:**
 - Development of Admin & Role-Based Dashboards for more tailored user experiences and system management.
 - Implementation of a Dark Mode Toggle for user preference and accessibility.
 - Addition of a Progress Bar on the Emergency Flow to keep users informed of the status.
- **System Quality Improvements:**
 - Incorporation of advanced Search & Filter Options within the application for better data retrieval and management.
 - Implementation of an After-Incident Notification system for updates and follow-ups.
 - Development of a Basic Analytics Module to provide insights into system usage and performance.

- **Addressing Domain Challenges:**
 - Further refinement of medical questionnaires with expert medical input.
 - Integration of real-world data for ambulance and hospital categorization and capabilities.
 - Enhancement of incident classification logic.
- **Integration of Optional Features:**
 - Full integration of Medical Records functionality once APIs are provided by Indus Hospital.
 - Population and integration of the Health Education section with provided resources.
- **Technical Enhancements:**
 - Exploring optimized or premium map services for better routing and visuals if resources allow.
 - Further optimization of backend services and database interactions for improved performance and scalability under load.

6. References

[1] Mukhtar, Sama, Syed Ghazanfar Saleem, Saima Ali, Sarfraz Ahmed Khatri, and Anna Q Yaffee. 2021. "Standing at the Edge of Mortality; Five-Year Audit of an emergency Department of a Tertiary Care Hospital in a Low Resource Setup". *Pakistan Journal of Medical Sciences* 37 (3).

[2] T. Alshugran and J. Dichter, "Extracting and modeling the privacy requirements from HIPAA for healthcare applications," *IEEE Long Island Systems, Applications and Technology (LISAT) Conference 2014*, Farmingdale, NY, USA, 2014, pp. 1-5, doi: 10.1109/LISAT.2014.6845198.

[3] K. Hjerpppe, J. Ruohonen and V. Leppänen, "The General Data Protection Regulation: Requirements, Architectures, and Constraints," *2019 IEEE 27th International Requirements Engineering Conference (RE)*, Jeju, Korea (South), 2019, pp. 265-275, doi: 10.1109/RE.2019.00036.